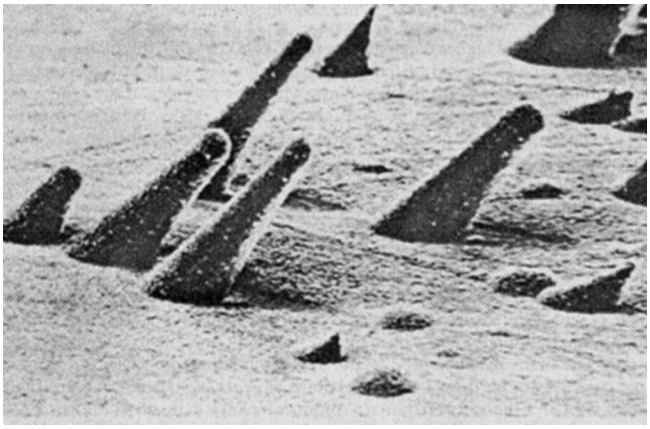


Figure 13.2 Holes in helmet of Apollo 12 astronaut caused by high energy cosmic rays. The holes have been made visible in a microscope by etching *(From New Scientist, April 22, 1973).*

Figure 13.2 shows the effect of high energy cosmic rays hitting the helmets of Apollo 12 astronauts. It is probable that the cosmic ray intensity will put a limit to how long man can endure in outer space: it has been calculated that in a journey to the planet Mars about 0.1% of the cerebral cortex will be destroyed. The annihilation process occurs to such an extent that below an altitude of approximately 25 km above the earth the number of primary cosmic particles has been reduced to quite a small fraction of the original intensity.

The discovery of π-mesons (or *pions*) was reported by Powell and Occhialini in 1948 after they had analyzed tracks in photographic emulsions placed for some months on a mountain top to get a high yield of cosmic ray interactions. Pions are produced in large amounts in all high energy ($\geq$400 MeV) nuclear reactions. In 1935 Yukawa suggested that the nucleons in a nuclide were held together through the exchange of a hypothetical particle, which we now recognize as the pion, just as hydrogen atoms in H_2 are held together through the exchange of an electron:

$$p \leftrightarrow n + \pi^+; \quad n \leftrightarrow p + \pi^-; \quad p \overset{\pi^\circ}{\leftrightarrow} p; \quad n \overset{\pi^\circ}{\leftrightarrow} n; \tag{13.1}$$

Pions are the particles of *strong interaction,* cf. §2.2. Since the pion rest mass is 0.147 u, the energy to produce a pion is at least 137 MeV (in practice, in order to conserve momentum it exceeds 400 MeV, cf. §4.6 and 5.7). The pions are unstable in free form.

13.2. SECONDARY REACTIONS IN THE EARTH'S ATMOSPHERE

Few of the pions formed in the annihilation process reach the earth's surface. They undergo radioactive decay (life-time about 10^{-6} s) to muons and muon neutrinos

(which will in a very short time undergo flavour oscillations) with a probability of about 99.99%. The second most probable fate of the pion is by decaying into an electron and a electron neutrino with a probability of about 0.01%. This latter decay was discovered in CERN in 1958. If none of the previous decay modes are realised they collide with other particles in the atmosphere and are annihilated. The muon is also an unstable subatomic particle with a mean lifetime for stationary muons of 2.2 μs. This comparatively long decay life time (the second longest known) is due to being mediated by the weak nuclear force. The only longer lifetime for an unstable subatomic particle is that for the free neutron which also decays via the weak force. All muons decay to three particles: an electron plus two neutrinos of different types.

The collision reactions of the pions result in the formation of a large number of other particles such as electrons, neutrons, protons, and photons. Some of the electrons so formed are captured in a thick zone around the earth known as the inner van Allen belt.

The main part (50–80%) of the cosmic radiation which reaches the earth's surface consists of high energy muons. Muons have much less tendency to react with atomic nuclei than pions and, therefore, can penetrate the atmosphere and the solid material of the earth relatively easily. The remaining part, which is the lower energy component of the cosmic radiation that strikes the earth, consists of photons, electrons, and positrons. At sea level this part of the cosmic radiation gives rise to approximately 2–3 ion pairs s^{-1} cm^{-3} of air. It is this component of the cosmic radiation that gives rise to the cosmic ray portion of the natural background that is measured by nuclear detection devices in laboratories.[1] Some of the cosmic radiation interacts to make atmospheric radioactivity which is principally the nuclides ^{3}H and ^{14}C.

13.3. RADIOELEMENTS IN NATURE

Practically all natural materials contain radioactive nuclides, though usually in such low concentrations that they can be detected only by very sensitive analysis. This is true for water (rain water, rivers, lakes, sea), rocks and soil, and all living matter, as well as for structures based on natural raw materials except where great care has resulted in use of a non-radioactive material. The radionuclides in our environment can be divided into (i) those formed from cosmic radiation, *cosmogenic*, (ii) those with lifetimes comparable to the age of the earth together with those that are part of the natural decay chains beginning with thorium and uranium, *primordial*, and (iv) those introduced into nature by man using modern techniques, *anthropogenic*.

[1] The remainder of the natural background comes from naturally occurring radioactive elements in the laboratory materials and surrounding building.